

Ritesh Somashekar

AI & ML Engineer

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PROFESSIONAL SUMMARY

- AI & Machine Learning Engineer with 6+ years of experience designing, deploying, and optimizing **scalable ML systems, NLP and LLM solutions**, and **production data pipelines** across healthcare and research domains. Proven expertise in **PyTorch, TensorFlow, and HuggingFace**, with hands-on experience in **LLM fine-tuning (LoRA, QAT)**, **agentic AI workflows**, and **low-latency model serving**.
- Strong background in **data engineering** with **Apache Spark, PySpark, Airflow, and Kafka**, and **cloud-based ML pipelines on AWS and Azure**, with a focus on **HIPAA-compliant** healthcare data. Experienced in **MLOps** using **MLflow, Docker, Kubernetes, FastAPI, and CI/CD pipelines**, enabling faster model iteration and reliable releases.
- Demonstrated impact, including **74% higher claims-data accuracy**, **60% faster model training**, and **70% fewer manual errors**, through **data-driven solutions, A/B testing, and model explainability (SHAP)**. Adept at partnering with cross-functional teams and client stakeholders to deliver **end-to-end ML solutions** from research to production.

TECHNICAL SKILLS

Programming & Scripting: Python (NumPy, pandas, scikit-learn, TensorFlow, PyTorch), SQL (PostgreSQL, Oracle), R, Scala, Java

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Deep Learning, Ensemble Models (XGBoost, LightGBM), Graph Analytics, Time Series Forecasting, Anomaly Detection, Feature Engineering, Hyperparameter Tuning (Optuna)

Natural Language Processing & LLMs: NLP, Transformers, BERT, BioBERT, Llama 3, GPT-4, SecureGPT, HuggingFace, Generative AI, Agentic AI, LoRA, Quantization-Aware Training, RAG, Prompt Engineering, Vector Embeddings

MLOps & ML Infrastructure: MLflow, Kubeflow, Airflow, Docker, Kubernetes, Jenkins, CI/CD, FastAPI, DVC, Model Deployment, Model Explainability (SHAP)

Data Engineering & Big Data: Apache Spark, PySpark, Kafka, Hadoop, Hive, Sqoop, ETL Pipelines, Delta Lake, Snowflake, MongoDB, Web Crawling (Selenium)

Cloud Platforms: AWS (S3, EC2, SageMaker, Lambda, Glue, Athena, SQS), Azure (Machine Learning, Stream Analytics, Data Services, Databricks)

Data Analysis & Visualization: Tableau, Power BI (DAX), Excel (VBA), Data Visualization, Statistical Analysis, A/B Testing

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA | GPA: 3.93/4.0

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India.

PROFESSIONAL EXPERIENCE

George Mason University | Graduate Research Assistant (Applied AI/ML) **Fairfax, VA | August 2024 - Present**

- Designed **Python** and **R** ML algorithms to study **social cohesion and community coping behavior during disasters** on large-scale real-world datasets, applying **graph-based analytics** and **ensemble models** on the **ORC Hopper HPC cluster** to forecast trends with an **RMSE of 0.89**.
- Developed a transformer-based NLP information-extraction pipeline using **BioBERT** and **HuggingFace Transformers**, processing **30K documents** and improving entity-recognition **F1-score by 18%** over baseline spaCy models.
- Fine-tuned **Llama 3 LLMs** using **LoRA** and **Quantization-Aware Training (QAT)**, reducing model size by **40%** while keeping accuracy within **3%** of full-precision benchmarks.
- Built an **agentic AI** ticket-automation system with **SecureGPT** and **LLaMA** on **AWS SQS** to monitor security events and generate JIRA tickets under NIST-CSF, reaching **93% automation accuracy** and cutting manual triage time by **70%**.
- Built an **anomaly detection** system using **LightGBM** with threshold-based alerting, reducing false positives by **25%**.
- Built end-to-end ML pipelines with **PyTorch, MLflow, and Optuna**, cutting model iteration cycles from **3 days to under 18 hours**, and standardized **15GB** of data across 6 datasets, reducing data-preparation time by **30%**.
- Deployed model inference APIs with **FastAPI** and **Docker** at **sub-200ms latency**, and streamlined MLflow-based releases from **2 weeks to 5 days** with **A/B testing** across model versions.

- Applied **SHAP** for model explainability, visualized findings in **R** and **Tableau**, and built reproducible workflows with **Git** and **DVC**, reducing onboarding time for new researchers to **under 2 days**.

Carelon Global Solutions | Machine Learning Engineer (Client: Accenture) Bengaluru, India | January 2019 – August 2023

- Designed and optimized **20+ ML-ready data pipelines** for provider and consumer analytics, focusing on **claims-balancing workflows** and scalable model-input generation using **Python, Apache Airflow**, and **AWS** with CI/CD.
- Developed Python ingestion workflows for **NPPES** provider data, improving claims and ML data accuracy by **74%** and reducing processing overhead.
- Built web crawlers that structured **400,000+ healthcare XML/JSON documents** for supervised learning, and engineered **SQL** and **DAX** feature pipelines that improved model training efficiency by **60%** and cut preprocessing overhead by **70%**.
- Designed NLP models using **BERT** and **BioBERT** for clinical entity extraction from unstructured medical text, achieving **80% accuracy** and reducing processing time by **55%** for a U.S. healthcare client.
- Automated **medical invoice classification** with rule-based scripting, reducing manual error rates by **70%**, and improved models with **Scikit-learn, XGBoost**, and **TensorFlow**, increasing accuracy by **8%** and cutting training time by **15%**.
- Built scalable **ETL pipelines** with **PySpark** and **AWS (S3, Glue, SageMaker)** on **Delta Lake**, improving data reliability by **30%** under **HIPAA**, and migrated legacy **Hive/Sqoop/Hadoop** jobs to **Apache Spark** and **Azure Data Services**.
- Orchestrated ingestion, feature-engineering, and model-retraining workflows with **Airflow DAGs**, reducing manual effort by **25%**.
- Built real-time analytics with **Azure Stream Analytics** and ML-powered **Tableau/Power BI** dashboards for anomaly detection and model-drift monitoring, reducing unplanned downtime by **20%**.
- Containerized ML inference services using **Docker, Kubernetes**, and **Jenkins CI/CD**, improving system uptime by **20%** and enabling real-time EHR integration via **REST APIs** secured with **OAuth 2.0**.
- Collaborated with Data Engineering, Product, and **Accenture** stakeholders on the **Seven Plus Locations** project to define ML use cases, KPIs, and roadmaps, reducing project turnaround time by **15%**.